Yuxin Xiao

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Research Interest

- **Understanding LLM Alignment**: Characterizing when and why language models fail to use information effectively, with a focus on safety under multi-turn and stylistic interactions, knowledge updates under retrieval augmentation, and intent drift in long-horizon multi-agent systems.
- **Improving LLM Alignment**: Developing model- and system-level frameworks for language model adaptation, spanning supervised fine-tuning for instruction following, reinforcement learning in healthcare, and agent harnesses for continual learning.

Education

Massachusetts Institute of Technology (MIT) — Cambridge, MA

Ph.D. in Social & Engineering Systems and Statistics — 08/2022 - 12/2026

- Advised by **Prof. Marzyeh Ghassemi**; GPA: 5.00/5.00
- Affiliated with Institute for Data, Systems, and Society (IDSS) and Laboratory for Information and Decision Systems (LIDS)

Carnegie Mellon University (CMU) — Pittsburgh, PA

M.S. in Machine Learning — 08/2020 - 12/2021

- Advised by **Prof. Eric Xing** and **Prof. Louis-Philippe Morency**; GPA: 4.12/4.33

University of Illinois at Urbana-Champaign (UIUC) — Urbana, IL

B.S. in Computer Science; B.S. in Statistics, Mathematics — 08/2016 - 05/2020

- Advised by **Prof. Jiawei Han** and **Prof. Hari Sundaram**; GPA: 3.93/4.00

Selected Publications

(First- or co-first-authored papers only; * indicates equal contribution; ordered by preprint date)

Recurrent Harness for Agentic Continual Learning — Preprint 2026
Yuxin Xiao, Zhiyun Lu, Jianyu Wang, Yangtian Zhang, Yang Song, Qifan Wang, Bailin Wang — [Paper], [Code]

Reinforcement Learning over Patient Trajectories for Clinical Reasoning in EHR Foundation Models — Preprint 2026
Yuxin Xiao, Sheng Zhang, Chandan Singh, Tristan Naumann, Hoifung Poon, Jianfeng Gao, Xiaodong Liu — [Paper], [Code]

When Style Breaks Safety: Defending LLMs Against Superficial Style Alignment — ICLR 2026
Yuxin Xiao, Sana Tonekaboni, Walter Gerych, Vinith Suriyakumar, Marzyeh Ghassemi — [Paper], [Code]

KScope: A Framework for Characterizing the Knowledge Status of Language Models — NeurIPS 2025
Yuxin Xiao, Shan Chen, Jack Gallifant, Danielle Bitterman, Thomas Hartvigsen, Marzyeh Ghassemi — [Paper], [Code]

Speak Easy: Eliciting Harmful Jailbreaks from LLMs with Simple Interactions — ICML 2025
Yik Siu Chan*, Narutatsu Ri*, **Yuxin Xiao***, Marzyeh Ghassemi — [Paper], [Code]

SFTMix: Elevating Language Model Instruction Tuning with Mixup Recipe — ACL 2026
Yuxin Xiao, Shujian Zhang, Wenxuan Zhou, Marzyeh Ghassemi, Sanqiang Zhao — [Paper], [Code]

In the Name of Fairness: Assessing the Bias in Clinical Record De-identification — FAccT 2023
Yuxin Xiao*, Shulammite Lim*, Tom Joseph Pollard, Marzyeh Ghassemi — [Paper], [Code]

Uncertainty Quantification with Pre-trained Language Models: A Large-Scale Empirical Analysis — EMNLP 2022
Yuxin Xiao, Paul Pu Liang, Umang Bhatt, Willie Neiswanger, Ruslan Salakhutdinov, Louis-Philippe Morency — [Paper], [Code]

SAIS: Supervising and Augmenting Intermediate Steps for Document-Level Relation Extraction — NAACL 2022
Yuxin Xiao, Zecheng Zhang, Yuning Mao, Carl Yang, Jiawei Han — [Paper], [Code]

Amortized Auto-Tuning: Cost-Efficient Bayesian Transfer Optimization for Hyperparameter Recommendation — Preprint 2021
Yuxin Xiao, Eric P. Xing, Willie Neiswanger — [Paper], [Code]

Heterogeneous Network Representation Learning: A Unified Framework with Survey and Benchmark — TKDE 2020
Carl Yang*, **Yuxin Xiao***, Yu Zhang*, Yizhou Sun, Jiawei Han — [Paper], [Code]

Discovering Strategic Behaviors for Collaborative Content-Production in Social Networks — WWW 2020
Yuxin Xiao, Adit Krishnan, Hari Sundaram — [Paper], [Code]

Non-local Attention Learning on Large Heterogeneous Information Networks — IEEE BigData 2019
Yuxin Xiao*, Zecheng Zhang*, Carl Yang, Chengxiang Zhai — [Paper], [Code]

Awards & Honors

2020 **C. W. Gear Outstanding Undergraduate Award**, UIUC (2 awardees at UIUC)

2020 **CRA Outstanding Undergraduate Researcher Award (Honorable Mention)**, CRA (135 awardees across North America)

Industry Research Experience

Meta MRS — Menlo Park, CA

Research Intern @ MRS Research Team — 06/2026 - 08/2026

Supervisor: **Dr. Bailin Wang**

Recurrent Harness for Agentic Continual Learning

- Proposed ReHarness, a recurrent harness framework for agentic continual learning that enables online adaptation under distribution shift through forward read/write and backward revision in a strictly streaming setting
- Designed a backward textual feedback mechanism that revises previously retrieved reflections using downstream trajectory feedback, allowing the harness state to correct outdated or overgeneralized experience without replay or validation data
- Demonstrated consistent gains across three continual learning benchmarks and three LLM backbones over raw-context, memory/reflection, and prompt-optimization baselines, while improving source-grounded revision, stability, and plasticity

Microsoft Research — Bellevue, WA

Research Intern @ Deep Learning Group & Health Futures — 06/2025 - 08/2025

Supervisor: **Dr. Xiaodong Liu**, **Dr. Chandan Singh**, **Dr. Sheng Zhang**, **Dr. Tristan Naumann**

Reinforcement Learning over Patient Trajectories for Clinical Reasoning in EHR Foundation Models

- Proposed a reinforcement learning fine-tuning framework that treats EHR foundation models as generative policies over longitudinal patient trajectories and formulates clinical prediction tasks as event-conditioned, time-windowed reasoning problems
- Designed time-aware, rollout-sensitive reward functions that account for finite generation horizons and temporally inconclusive outcomes in longitudinal EHR data
- Demonstrated that RL fine-tuning consistently improves over pre-trained backbones and strong baselines, enabling smaller models to surpass larger ones in data-limited regimes, inducing positive cross-task transfer, and improving structural, semantic, and downstream predictive utility

Zoom AI — San Jose, CA

Research Intern @ GenAI R&D Team — 06/2024 - 08/2024

Supervisor: **Dr. Shujian Zhang**, **Dr. Sanqiang Zhao**

SFTMix: Elevating Language Model Instruction Tuning with Mixup Recipe

- Designed and implemented SFTMix, a novel Mixup-based instruction-tuning algorithm that leverages training dynamics to identify confidence-based data subspaces and improve large language models' instruction tuning without curated datasets
- Conducted extensive experiments demonstrating that SFTMix consistently outperforms standard next-token prediction (NTP) across diverse instruction-following and healthcare-specific benchmarks
- Performed in-depth analysis of data confidence, regularization, and scalability, showing SFTMix's compatibility with data selection, parameter-efficient fine-tuning (LoRA), and compute-constrained deployment scenarios

Bosch Center for Artificial Intelligence — Stuttgart, Germany

Research Intern @ NLP Team — 06/2023 - 08/2023

Supervisor: **Prof. Annemarie friedrich**, **Prof. Heike Adel**

InstructTransfer: Transfer Tuning of Human-Language Instructions for Black-Box Large Language Models

- Developed an automatic and cost-effective instruction-tuning strategy for black-box language models (e.g., ChatGPT) by transferring knowledge from prior tuning tasks via an optimal-transport-based task similarity kernel
- Outperformed existing tuning baselines on downstream natural language generation applications with only one-eighth of the tuning cost

Academic Research Experience

Healthy ML Group — MIT

Supervisor: **Prof. Marzyeh Ghassemi** — 08/2022 - Present

When Style Breaks Safety: Defending Language Models Against Superficial Style Alignment

- Investigated how style patterns compromise LLM safety by defining and quantifying ASR inflation across 36 models and seven benchmarks
- Characterized the mechanisms behind increased vulnerability, demonstrating that ASR inflation correlates with models' relative attention to style patterns and their overlap with instruction-tuning data, and that superficial style alignment further amplifies jailbreak susceptibility
- Developed and validated SafeStyle, a defense strategy that augments a small amount of safety training data with matched style distributions, achieving consistent safety improvements across three models, six fine-tuning style settings, and two real-world instruction-tuning datasets

KScope: A Framework for Characterizing the Knowledge Status of Language Models

- Proposed a five-status taxonomy of LLM knowledge based on the consistency and correctness of model responses, going beyond traditional conflict-based evaluations to characterize what a model knows about a given question more precisely
- Developed KScope, a hierarchical statistical testing framework that progressively refines hypotheses about LLM knowledge modes and assigns one of the five knowledge statuses
- Conducted large-scale evaluations across nine LLMs and four datasets, revealing key insights into how supporting context narrows knowledge gaps, how context features influence knowledge updates, and how targeted context summarization improves update effectiveness

Speak Easy: Eliciting Harmful Jailbreaks from LLMs with Simple Interactions

- Investigated the real-world impact of jailbreak vulnerabilities by evaluating whether LLM responses are sufficiently actionable and informative to enable average users to carry out harmful actions
- Proposed HarmScore, a novel evaluation metric that quantifies how effectively jailbreak responses facilitate harmful behavior, offering a more practical perspective on LLM safety beyond traditional success rate metrics
- Developed Speak Easy, a multi-step, multilingual attack framework that exploits common interaction patterns, leading to substantial increases in jailbreak effectiveness and revealing critical, overlooked vulnerabilities in existing safety evaluations

In the Name of Fairness: Assessing the Bias in Clinical Record De-identification

- Examined the bias in de-identifying clinical discharge notes for patients of various demographic groups by conducting a large-scale empirical evaluation of nine popular de-identification baseline methods of different categories
- Prepared and annotated 100 note templates based on real-world clinical discharge notes and 16 name lists that were representative of different gender, race, name popularity, and the decade of popularity
- Investigated three factors that affected the de-identification quality and proposed an effective and method-agnostic solution by fine-tuning de-identification methods with clinical context and diverse names

MultiComp Lab — CMU

Supervisor: **Prof. Louis-Philippe Morency** — 04/2021 - 04/2022

Uncertainty Quantification of Pre-trained Language Models: A Large-Scale Empirical Analysis

- Presented a holistic empirical analysis on how to compose a well-calibrated pre-trained language model-based prediction pipeline
- Examined how different pre-trained language models worked with various uncertainty quantifiers under distribution shifts and on diverse tasks
- Investigated the influence of pre-training datasets and strategies, model sizes, fine-tuning strategies, and uncertainty quantifiers on the calibration quality of pre-trained language models based on thousands of empirical observations
- Inspected the relationship between calibration quality and other aspects of model performance such as accuracy, sharpness, and robustness

SAILING Lab — CMU

Supervisor: **Prof. Eric Xing** — 09/2020 - 12/2021

Amortized Auto-Tuning: Cost-Efficient Bayesian Transfer Optimization for Hyperparameter Recommendation

- Proposed a multi-task multi-fidelity Bayesian optimization framework with a novel task kernel and acquisition function. Leveraged cheap-to-obtain low-fidelity observations to efficiently recognize promising hyperparameters for new tuning tasks via knowledge transfer
- Analyzed the cost-efficiency and flexibility of existing hyperparameter tuning baselines by surveying 36 methods based on seven specific criteria
- Computed a hyperparameter recommendation database to serve the research communities, which consisted of 27 unique image classification tuning tasks and 150 distinct configurations over a 16-dimensional nested hyperparameter space

Data Mining Group — UIUC

Supervisor: **Prof. Jiawei Han** — 03/2019 - 05/2021

SAIS: Supervising and Augmenting Intermediate Steps for Document-Level Relation Extraction

- Developed a multi-task learning framework for document-level relation extraction, which consisted of a broad spectrum of carefully designed tasks to explicitly teach pre-trained language models to capture the key sources of information—relevant contexts and entity types
- Boosted the model performance further via evidence-based data augmentation and ensemble inference, while preserving the computational cost by assessing the uncertainty of model predictions
- Achieved state-of-the-art relation extraction results on three benchmarks and outperformed the runner-up by 5.04% relatively in retrieving interpretable evidence for each extracted relation

Heterogeneous Network Representation Learning: A Unified Framework with Survey and Benchmark

- Analyzed and evaluated the performance of various types of heterogeneous network embedding models (proximity-preserving based, message-passing based, relation-learning based) on different applications (node classification, link prediction)
- Designed and implemented a unified and user-friendly experiment interface for fair and efficient comparison of 13 popular heterogeneous network embedding algorithms on four large-scale benchmark datasets; received over 300 stars and forks on the corresponding GitHub repository

Crowd Dynamics Lab — UIUC

Supervisor: **Prof. Hari Sundaram** — 10/2018 - 04/2020

Discovering Strategic Behaviors for Collaborative Content-Production in Social Networks

- Investigated the research question of whether resource-limited individuals were able to discover strategic behaviors associated with high payoffs when producing collaborative/interactive content in social networks
- Proposed a novel dynamic dual graph attention network which modeled individuals' content production strategies under social influence through a generative process
- Conducted a thorough qualitative analysis on a real-world social network with over seven million nodes and 400 million edges, which revealed three strong empirical findings about the emergence of individuals' strategies during the content production stage

Teaching Experience

Massachusetts Institute of Technology (MIT) — Cambridge, MA

Head Teaching Assistant & Session Instructor

6.S977 Ethical Machine Learning in Human Deployments, Instructors: *Prof. Marzyeh Ghassemi* — 02/2026 - 05/2026
14.310x Data Science for Social Scientists, Instructors: *Prof. Esther Duflo*, *Prof. Sara Ellison*; Coordinator: *Dr. Karene Chu* — 06/2022 - 08/2022

University of Illinois at Urbana-Champaign (UIUC) — Urbana, IL

Teaching Assistant

CS446 Machine Learning, Instructor: *Prof. Sanmi Koyejo* — 08/2019 - 12/2019
CS410 Text Information Systems, Instructor: *Prof. Chengxiang Zhai* — 08/2019 - 12/2019
CS125 Intro to Computer Science, Instructor: *Prof. William Chapman* — 01/2017 - 05/2017

Academic Service

- **Workshop Organizer**: LLM/VLM Deployment Opportunities and Risks in Healthcare @ COLM 2026
- **Conference & Journal Reviewer**: NeurIPS, ICML, ICLR, ACL Rolling Review, CHIL, ML4H, TMLR, JAMIA, Nature Medicine, Nature Human Behaviour
- **Research Mentor**: Yik Siu Chan (now PhD student at UMD)